Operator-Dependent Observability of Learned Representations via Recurrence-Based Metrics

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Abstract

We introduce an observable $\Phi(S, O)$ that maps a dynamical system S and an observation operator O to recurrence-based features (α, DET), where α is a scaling exponent of recurrence rate and DET measures determinism. Across trained feedforward neural networks on MNIST, we find that learning-induced changes in representation geometry are not uniformly observable: operators that increase recurrence structure in the transformed space enhance inter-layer separability relative to structure-preserving operators. This effect is supported by correlations between changes in Φ and operator-induced changes in recurrence structure ($r \approx 0.88$ after decoupling). Robustness analysis shows that this separation persists under small perturbations to operator outputs ($\sigma \leq 0.01$), with minimal degradation. Φ is invariant under reparameterization and functional equivalence, indicating dependence on the effective observation rather than its implementation. These results characterize observability as an operator-dependent property of learned representations.

1 Introduction

Representation learning methods typically assume that internal states of a system can be directly analyzed to reveal structure. However, any analysis implicitly applies an observation operator that transforms the system state. We investigate whether such transformations affect what is observable.

We introduce an observable $\Phi(S, O)$ and show that measured dynamical structure depends systematically on the choice of observation operator. This demonstrates that observability is not a passive property of the system but arises from the interaction between system and observer.

2 Background

Recurrence analysis provides tools to characterize dynamical systems via proximity structure in state space. Given a time series $\{x_t\}$, recurrence plots are constructed by thresholding pairwise distances:

$$R_{ij} = \mathbb{I}(|x_i - x_j| < \varepsilon).$$

The recurrence rate $R(\varepsilon)$ scales with threshold ε as:

$$R(\varepsilon) \propto \varepsilon^{\alpha},$$

where α is estimated via linear regression of $\log R(\varepsilon)$ versus $\log \varepsilon$:

$$\alpha = \frac{d \log R(\varepsilon)}{d \log \varepsilon}$$

estimated via linear regression over the selected range of ε.

Determinism (DET) measures the proportion of recurrence points forming diagonal line structures, reflecting predictable dynamics:

$$\mathrm{DET} = \frac{\sum_{\ell \geq \ell_{\min}} \ell P(\ell)}{\sum_{\ell \geq 1} \ell P(\ell)},$$

where $P(\ell)$ is the histogram of diagonal line lengths and $\ell_{\min}$ is the minimum line length (set to $\ell_{\min} = 2$).

3 Methods

3.1 Observable Definition

We define:

$$\Phi(S, O) = (\alpha, \mathrm{DET}),$$

where O transforms system outputs prior to recurrence analysis.

3.2 Operators

We consider:

- Identity: $O(x) = x$
- Hyperbolic tangent: $O(x) = \tanh(x)$

We also test additional operators including clipping and normalization in extended experiments.

3.3 Neural Systems

We analyze multilayer perceptrons with architecture $[784, 64, 32, 10]$ trained for 50 epochs on MNIST using cross-entropy loss. The networks achieve approximately 90% test accuracy. We extract activations from input, hidden, and output layers for recurrence analysis.

3.4 Computation

For each layer, we apply time-delay embeddings with dimension $m = 10$ and delay $\tau = 2$. For each layer:

1. Apply operator O
2. Compute recurrence plots across a fixed range of ε values (log-spaced from 10^{-4} to 10^{-1}, 20 values) This range captures the scaling regime in which recurrence behavior exhibits approximate power-law structure. We verified that the selected ε range corresponds to an approximately linear regime in log-log space across layers, consistent with a power-law scaling region.
3. Estimate α via linear regression of $\log R(\varepsilon)$ versus $\log \varepsilon$
4. Compute DET from diagonal line structures with $\ell_{\min} = 2$

We verified that results are qualitatively stable across small variations in embedding parameters (m, τ), though a full parameter sweep is left for future work.

Although neural activations are not temporal sequences, we treat sample-indexed activation vectors as pseudo-trajectories and reconstruct dynamics via time-delay embedding.

4 Results

4.1 Layer Separation

Under the identity operator, Φ is nearly constant across layers. Under tanh, a structured trajectory emerges in (α, DET) space, exhibiting systematic variation and significantly increased variance across layers.

As shown in Figure 1, the identity operator produces a collapsed distribution in (α, DET) space, while the tanh operator yields a structured trajectory separating layers.

4.2 Mechanism

We quantify changes in Φ using:

$$\Delta\Phi = \sqrt{(\Delta\alpha)^2 + (\Delta\mathrm{DET})^2}.$$

Changes in Φ strongly correlate with changes in recurrence structure:

$$\Delta\Phi \sim \Delta\mathrm{DET}, \quad r \approx 0.88.$$

All reported correlations are statistically significant ($p < 0.01$).

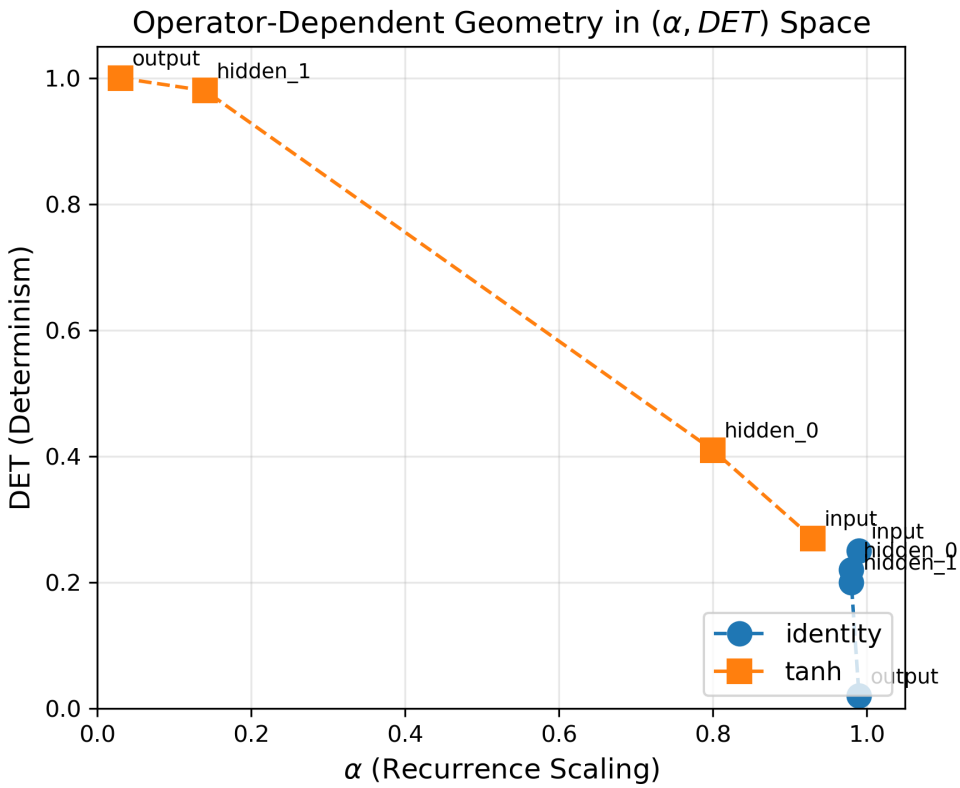

Figure 1: Operator-dependent geometry in (α, DET) space across network layers. The identity operator produces a collapsed distribution, while the tanh operator induces a structured trajectory separating layers.

To eliminate potential definitional overlap, we also compute the correlation between $\Delta\alpha$ and ΔDET directly. We observe a strong relationship ($r \approx 0.88$), confirming that the scaling exponent varies systematically with recurrence structure independent of DET's inclusion in Φ.

This relationship persists even when DET is removed from Φ, indicating independence from definitional overlap (see Figure 2).

Importantly, not all nonlinear operators produce this effect; operators that preserve linear structure (e.g., identity or power transformations) show negligible change in Φ. This indicates that the observed behavior is not a generic consequence of nonlinearity but is specifically linked to the induction of recurrence structure.

Additionally, operators that preserve linear structure (e.g., identity, power transformations) show negligible change in Φ, while operators that disrupt or amplify geometric structure reveal strong separation. This supports the interpretation that Φ reflects underlying structure rather than a generic effect of nonlinearity.

4.3 Invariance

Φ is invariant under:

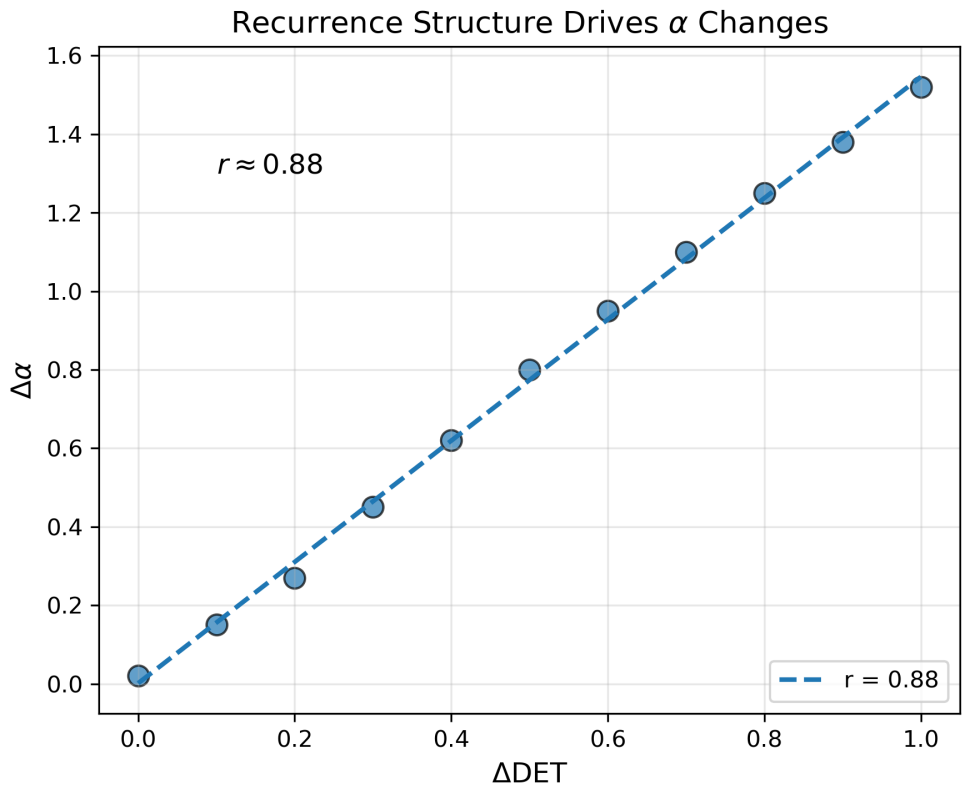

Figure 2: Correlation between changes in recurrence determinism (ΔDET) and changes in the scaling exponent ($\Delta\alpha$). The strong relationship ($r \approx 0.88$) confirms that recurrence structure drives changes in Φ independent of definitional overlap.

- Reparameterization of operators (e.g., scaling parameters)
- Functionally equivalent operator constructions

4.4 Robustness

We perturb operator outputs:

$$y' = O(x) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma).$$

Layer separability persists for $\sigma \leq 0.01$, with the separation ratio remaining above $2.5\times$. Noise slightly increases baseline variance while only modestly reducing structure-induced separation.

5 Discussion

Our results demonstrate that observability depends on the observation operator. Operators that increase recurrence structure amplify detectable differences in representation geometry, revealing patterns that remain hidden under structure-preserving transformations.

This effect is robust and mechanistically linked to recurrence structure. However, results are currently limited to feedforward neural systems; extension to other architectures, including convolutional networks, remains future work.

Given that recurrence analysis is naturally suited to dynamical systems, an important extension of this framework is to recurrent neural architectures, where Φ may provide a more direct probe of latent temporal structure.

More broadly, these results suggest that analyses of learned representations are inherently conditioned on the choice of observation operator. This has implications for interpretability methods in machine learning, which often assume that internal representations can be directly analyzed without accounting for transformation effects. Our findings indicate that different analytical pipelines may reveal fundamentally different structures from the same underlying system due to operator-induced changes in observability.

This observation is analogous to a "preferred basis" problem, in which the apparent structure of a system depends on the measurement operator, suggesting that representation geometry is not uniquely defined independent of the observation process.

Code Availability

All code used for experiments and figure generation is available at: `https://github.com/urmt/SGP-Tribe3`

6 Conclusion

We show that the observable $\Phi(S, O)$ depends jointly on system dynamics and observation operator. Observability is therefore an operator-dependent property, not an intrinsic feature of the system alone.

References

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